

SOC Investigation Report #5

PowerShell Activity Investigation (Sysmon Event ID 1)

Objective

Investigate PowerShell activity using Sysmon Process Creation events. Analyze the parent-child process relationship, executed commands, user account, and determine whether the observed activity is legitimate or suspicious.

Lab Environment

Component	Details
Host	Windows 11
Virtualization	Oracle VirtualBox
Victim	Windows 10 Home
Attacker	Kali Linux
SIEM	Splunk Enterprise
Forwarder	Splunk Universal Forwarder
Monitoring	Sysmon v15.21

Commands Executed

PowerShell:
whoami
hostname
Get-Date
Get-ChildItem C:\
Get-Process

Splunk Query

```
index=* EventCode=1  
| search User="WINDOWS\\biru"  
| table _time Image CommandLine User ParentImage
```

Observed Evidence

Image: C:\Windows\System32\whoami.exe
Image: C:\Windows\System32\HOSTNAME.EXE

Parent Image:
C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

User:
WINDOWS\biru

Investigation

Sysmon recorded the execution of whoami.exe and hostname.exe. Both processes were launched from powershell.exe by the logged-in user. The parent-child relationship confirms the commands were executed from an interactive PowerShell session.

SOC Analysis

Observed commands (whoami and hostname) are common administrative commands used to identify the current user and computer name. No encoded commands, downloads, or suspicious child processes were observed.

Conclusion

This activity is legitimate administrative PowerShell usage. No indicators of compromise were identified. No security incident should be raised based on this evidence.

Evidence

Insert screenshots:

1. PowerShell window showing executed commands.
2. Splunk search results.
3. Event highlighting Image, ParentImage, User and CommandLine.

Skills Learned

- Process Creation (Sysmon Event ID 1)
- Parent-child process analysis
- PowerShell investigation
- Splunk event analysis
- Distinguishing legitimate vs suspicious PowerShell activity